

Forecast Models Analysis and Accuracy Assessment with Realtime Observed Data

A report submitted to Pakistan Meteorological Department in partial fulfillment of internship criteria

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Acknowledgments

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Abstract

Summer Monsoon is a huge climate system that commences from the month of July till the month of September. It is responsible for 65 to 75 percent of the annual water in Pakistan, thereby playing a pivotal role in the Agriculture sector and in the replenishment of the country's water resources. Due to summer monsoon rainfall, Pakistan is more prone to floods and in 2022 Pakistan has received 190 percent of its normal rainfall which is a record breaking amount. This year the precipitation in July has exceeded the normal monsoon rainfall by 26 percent, making it the wettest month since 1961.

With the help of the forecasts models we have used along with the real-time data, we can carry out evaluation of precipitation for the month of July 2022, and determine which areas are most likely to be impacted by heavy rainfall.

Refer to Figure 1.0 that outlines the average rainfall per hour (mm/hr) for each province of Pakistan and its analysis with rainfall thresholds as set by United States Geological Survey (USGS),

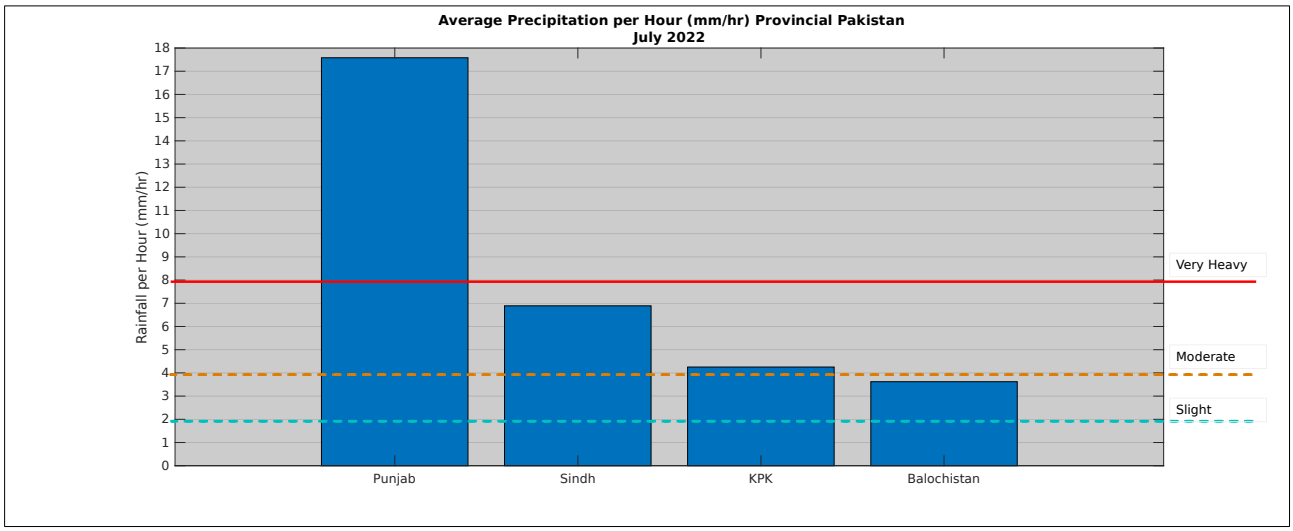


Figure 1.0, Provincial Rainfall per hour (mm/hr) with USGS thresholds

Overview

This report summarizes the internship progress made by the respective authors as internees at Pakistan Meteorological Department (PMD). While primarily covering the entire tenure, the report will focus on the evaluation of precipitation forecasts with real-time data for the month of July 2022, Pakistan. The forecasts models used are as follows European Center for Medium-Range Weather Forecasts (ECMWF), Global Forecasting System (GFS), and Icosahedral Non-hydrostatic (ICON) while real-time data is provided by PMD.

1 Forecast Models and Pre-processing

Using Ubuntu OS from a native Linux Distribution, Grid Analysis Display System (GrAds) and Climate Data Operator (CDO) were used to evaluate the models deployed from a command line interface. The forecasts models along with realtime observed data were provided by the host Institution PMD.

The data models are summarized as below:

-using NetCDF command **ncdump -h <filename>** a metadata descriptor was generated

1. ECMWF
    1. Large Scale Precipitation (meters)
    2. Convective Precipitation (meters)
       -6 hour timestep data
       -Commulative data

2. GFS
    1. Nonacumulated Precipitation (mm)
       -6 hour timestep data

3. ICON
    1. Total Precipitation (mm)
       -3 hour timestep data
       -Commulative data

All data provided was set on a global scale with latitude -90° to 90° and longitude -180° to 180°, therefore the data sets had to be masked according to the following political boundaries;

1. All provinces of Pakistan separately
2. Entire Pakistan inclusive of all provinces
   *refer to Figure 1.1 as reference



Figure 1.2.1 Boundaries of Pakistan

For the purpose, CDO commands were initiated are each output file store separately. The data was further sorted into weeks starting from 0300 hours of the first day of each week onward the seventh day.

*0300 hours is the beginning of forecast projections hence it was wise to begin each day with 0300 hours instead of 0000 hours.

2 Results and Analysis

2.1 Post-processing

To obtain a better estimate of the values each obtained value was interpolated with its area weight. Since we are dealing with geospatial data, the area weight is a spatial weight matrix representation of the spatial structure of your data. It is a quantification of the spatial relationships that exist among the features in a dataset. Hence the average topographic estimate of an area.

2.2 Graphical Analysis of Forecast Models with Observed Data

Using GrAds and QGIS GUI Software average precipitation data was plotted on a feature map of Pakistan for a better geospatial variation and visualization.

Weekly Average Precipitation Distribution of Pakistan July 2022

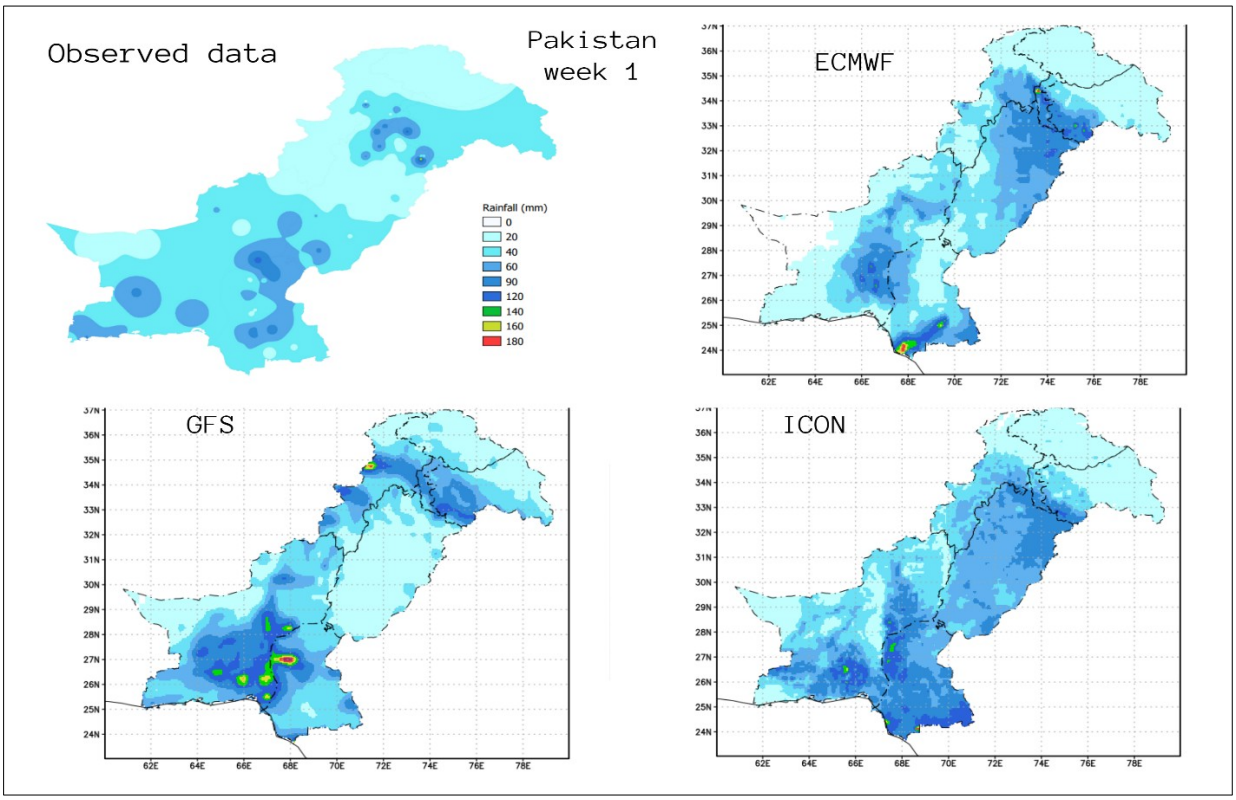


Figure 2.2.1, Average Precipitation Pakistan Week 1 July 2022

From **Fig 2.2.1** it can deduced that Gilgit Baltistan experienced the least amount of rainfall in the 1st week, with average rainfall falling between 20-40 mm. It can be seen from all four maps displayed above that lower Sindh and south-eastern parts of Balochistan experienced heavy average rainfall of 90-120 mm while according to ECMWF model some places in extreme south of Sindh has received 140 -180 mm of rainfall. Some of the areas in southern Sindh and east-southern Balochistan that

can experience urban flooding if such level of precipitation persists are Karachi, Hyderabad, Thatta, Dadu, Larkana, Jacobabad, Khuzdar and Sukkur etc.

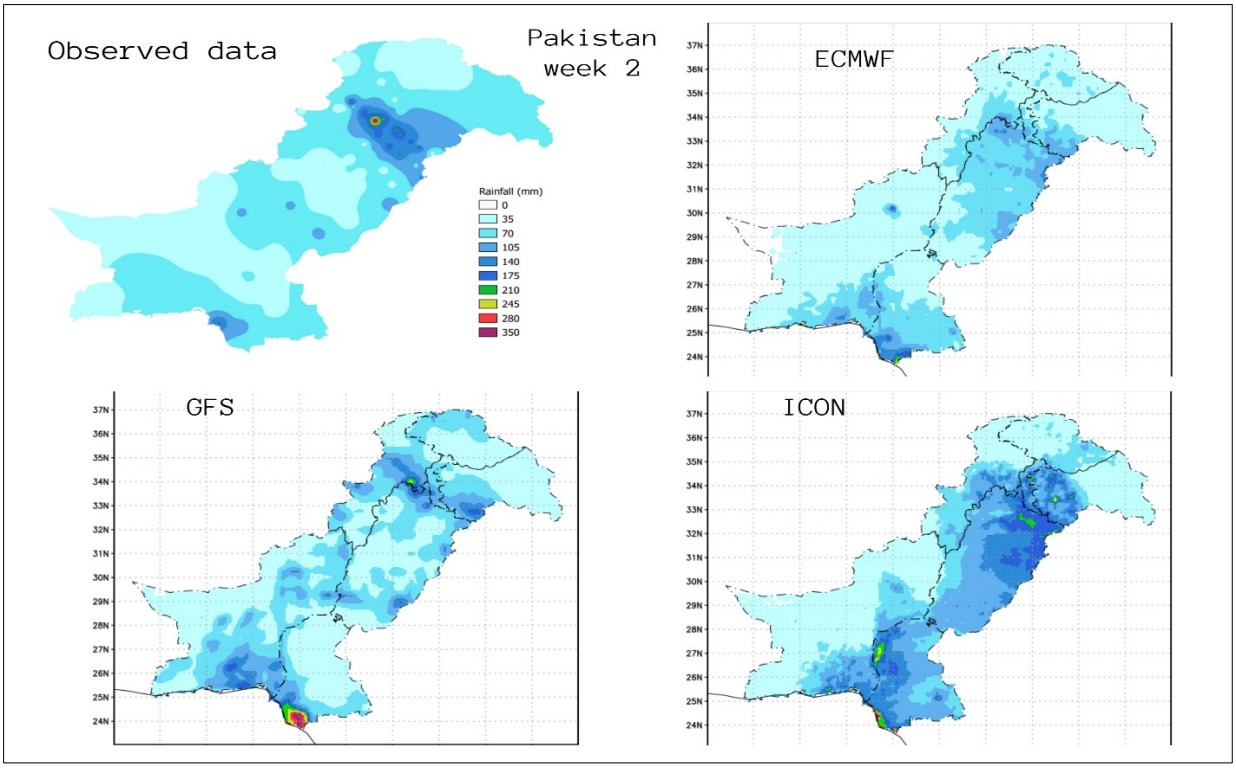


Figure 2.2.2, Average Precipitation Pakistan Week 2 July 2022

From **Fig 2.2.2** we can see that Punjab, federal capital and north eastern parts of KPK has received rainfall of 70-140 mm. Some areas in Punjab have received rainfall up to 170 mm.

Extreme south of Sindh continues to receive rainfall with the upper average rainfall threshold of 175 mm in observed data and ECMWF model, and with an upper average threshold of 210-280 mm in GFS and ICON model specifically in areas located in the extreme south of Sindh.

Moreover, Observed data and ECMWF model show that most parts of Balochistan and Sindh received rainfall of 35-70 mm whereas only few areas mainly located in southern regions received average rainfall of around 175 mm. In GFS and ICON model we can see that comparatively larger number of areas have experiences average rainfall of 105 -175 mm.

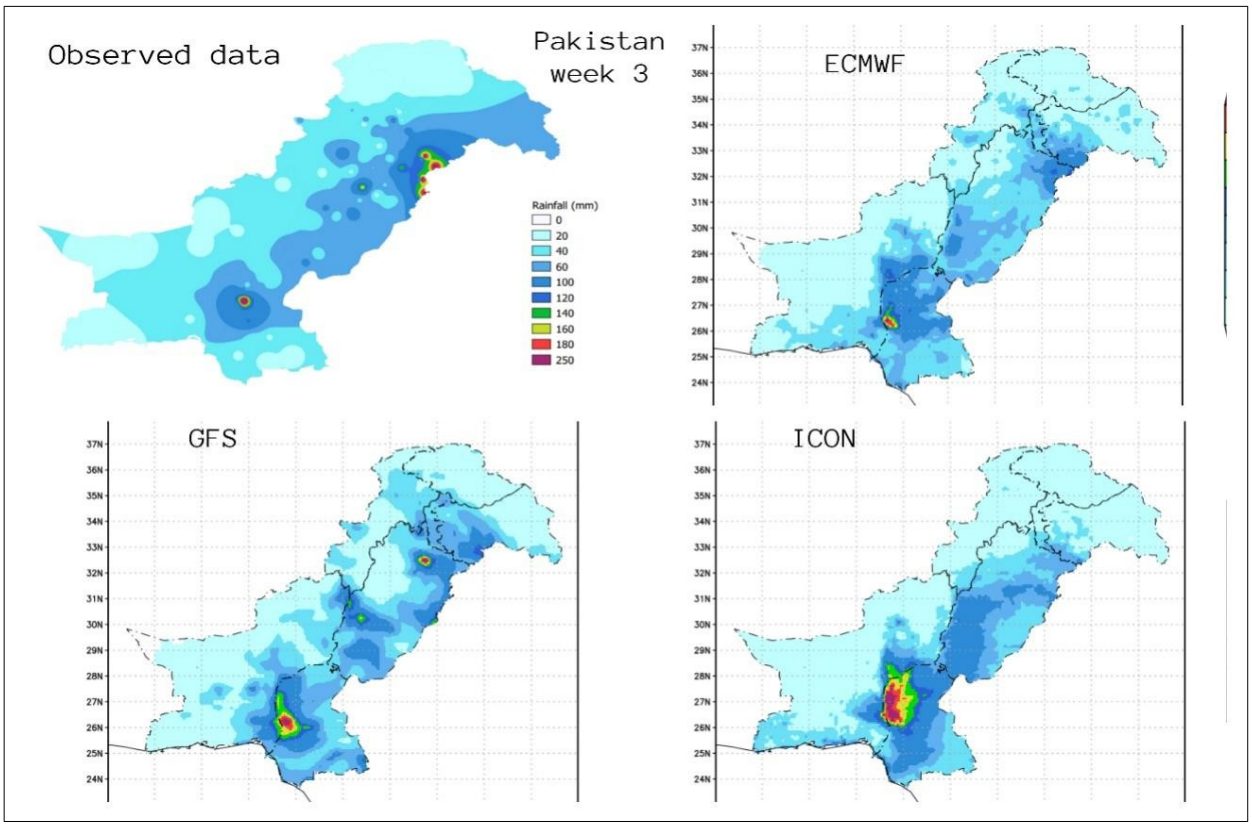


Figure 2.2.3, Average Precipitation Pakistan Week 3 July 2022

From **Fig 2.2.3**, it can be seen that Sindh and south-eastern parts of Balochistan along with northern parts of Punjab and some parts of KPK have received ample amount of rainfall in the third week of July. Western Sindh along with north eastern Baluchistan continues to receive heavy rainfall with the upper threshold reaching 120 mm in east-southern parts of Balochistan and 250 mm in western Sindh. Observed data and ECMWF model show approximately similar area coverage of those regions that received very heavy rainfall in western Sindh whereas GFS and ICON model slightly differ from observed data as greater number of areas fall under those that received very heavy rainfall. Furthermore, Gilgit Baltistan and most of KPK continues to receive average rainfall in the range of 20-40 mm with there being very few parts that received 60 mm.

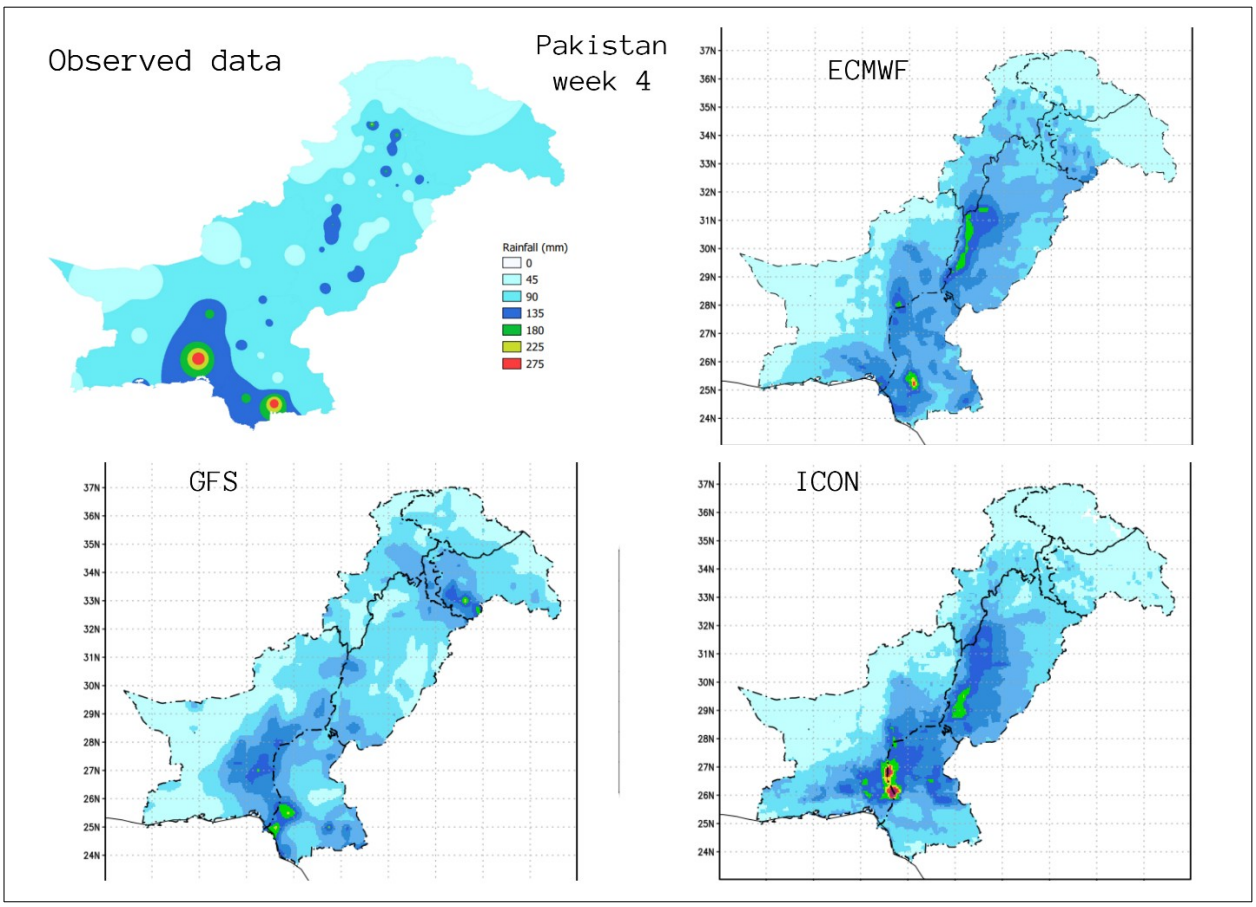


Figure 2.2.4, Average Precipitation Pakistan Week 4 July 2022

In **Fig 2.2.4**, it is evident that south-eastern Balochistan and Sindh continues to receive heavy rainfall in the 4th week of July with upper thresholds reaching 275 mm of average rainfall. Western Punjab also received heavy rainfall of around 135 -180 mm.

Moreover, all models discussed are reliable but as mentioned before, ECMWF appears to be the most similar to the observed data comparatively.

2.3 Non-Graphical Statistical Analysis of Forecast Models with Observed Data

The obtained forecast values were then analyzed in a side by side comparison with the real-time values being the accuracy assessment criteria. The results saved in **.csv** file were plotted in MATLAB Software using OpenGL rendering.

Comparison of Models based on daily average forecasts for Pakistan July 2022

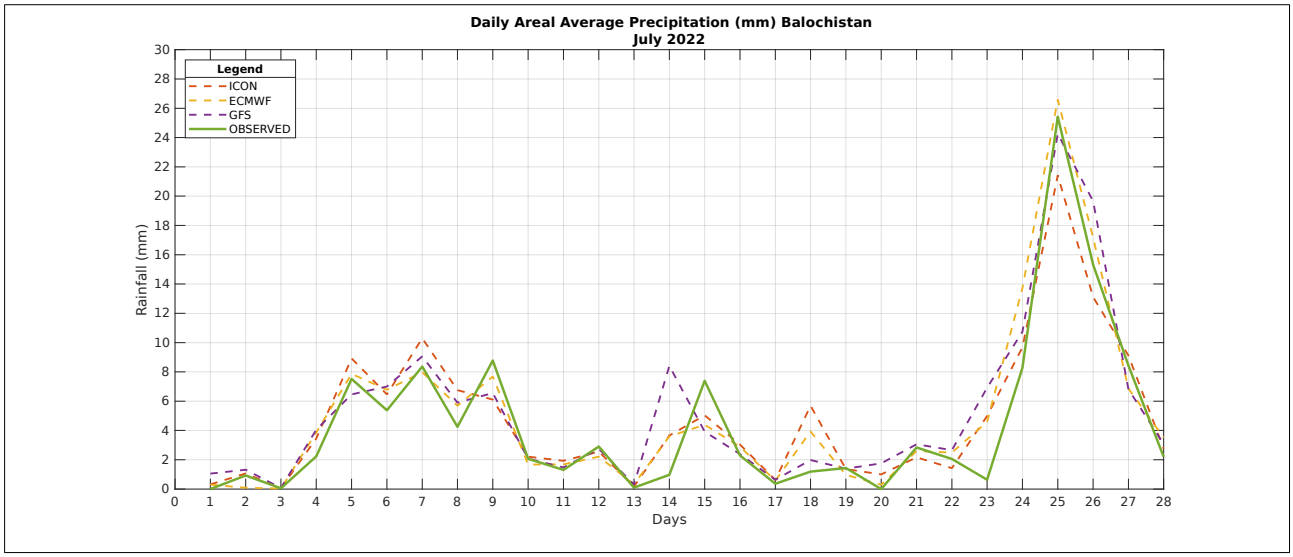


Figure 2.3.1, Daily Areal Average Precipitation Balochistan, Pakistan July 2022

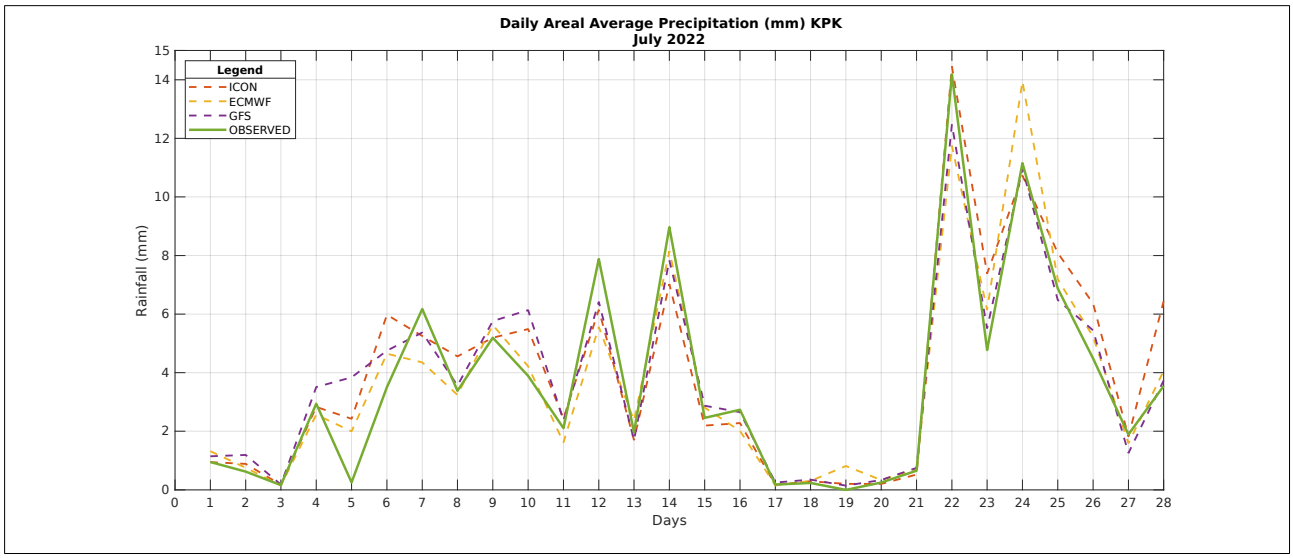


Figure 2.3.2, Daily Areal Average Precipitation KPK, Pakistan July 2022

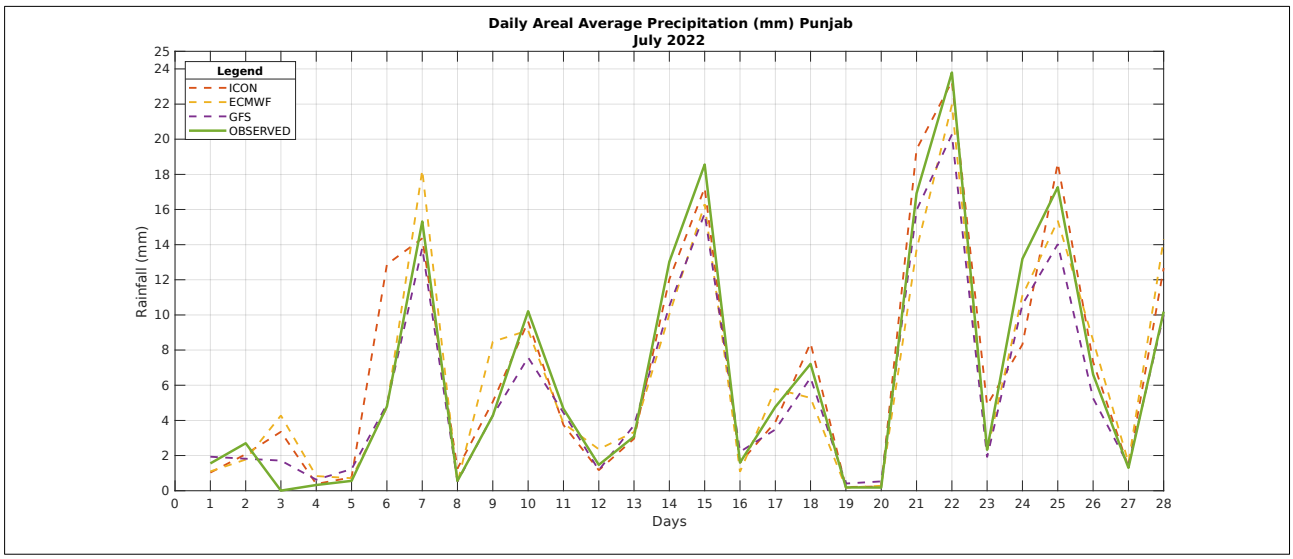


Figure 2.3.3, Daily Areal Average Precipitation Punjab, Pakistan July 2022

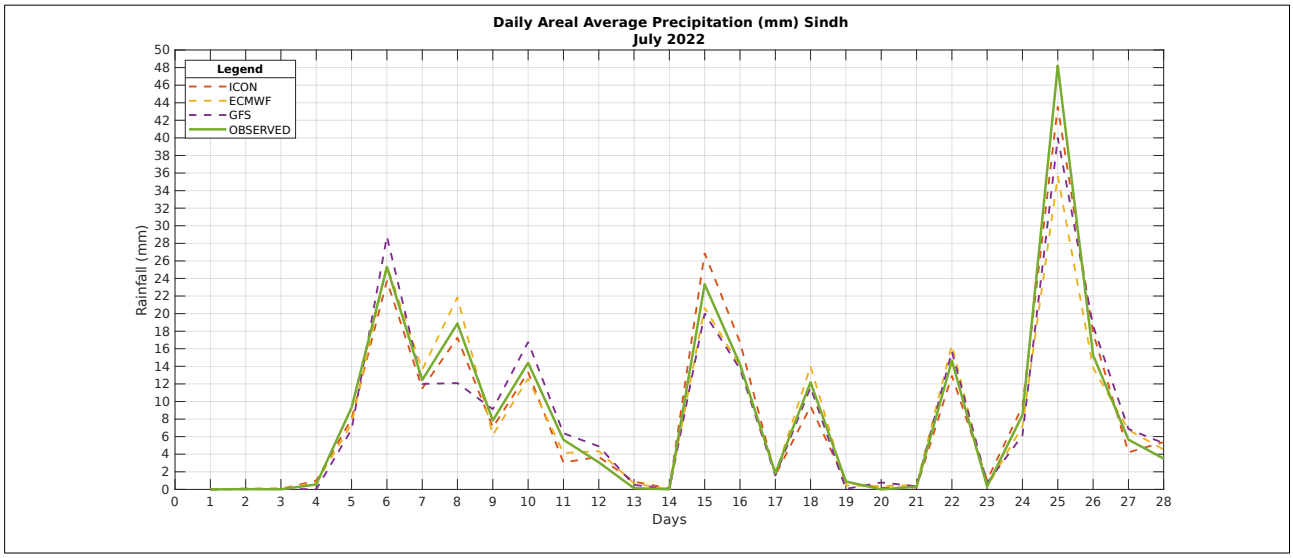


Figure 2.3.4, Daily Areal Average Precipitation Sindh, Pakistan July 2022

The above statistical visualization shows that all forecast models agree well with the observed data, with ECMWF forecast having the greatest similarity.

Comparison of Models based on weekly average forecasts for Pakistan July 2022

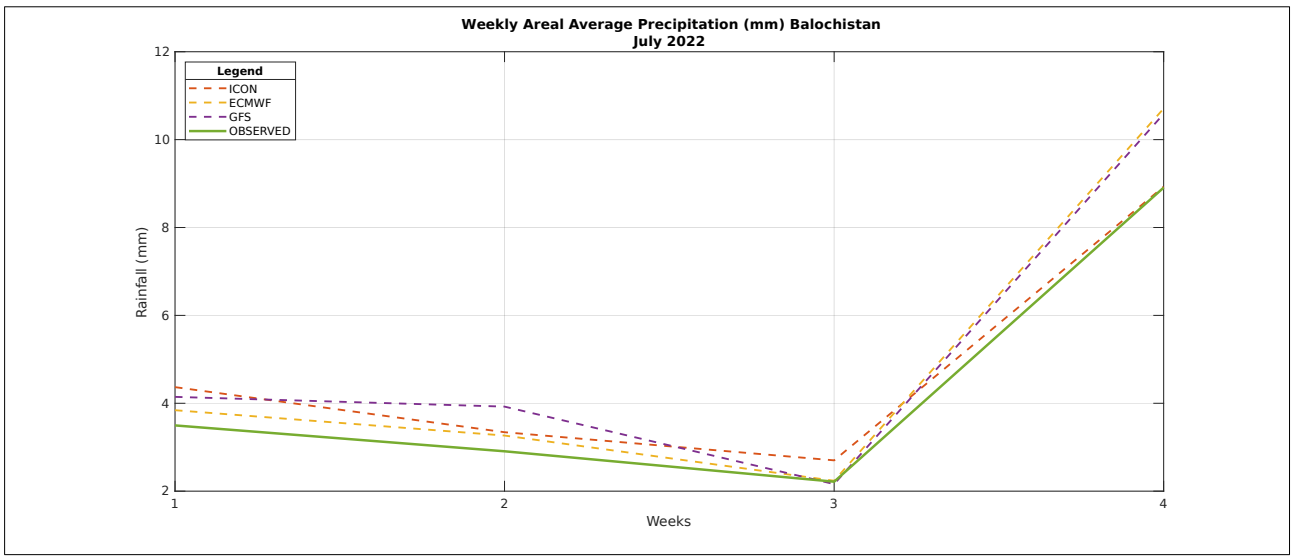


Figure 2.3.5, Weekly Areal Average Precipitation Balochistan, Pakistan July 2022



Figure 2.3.6, Weekly Areal Average Precipitation KPK, Pakistan July 2022

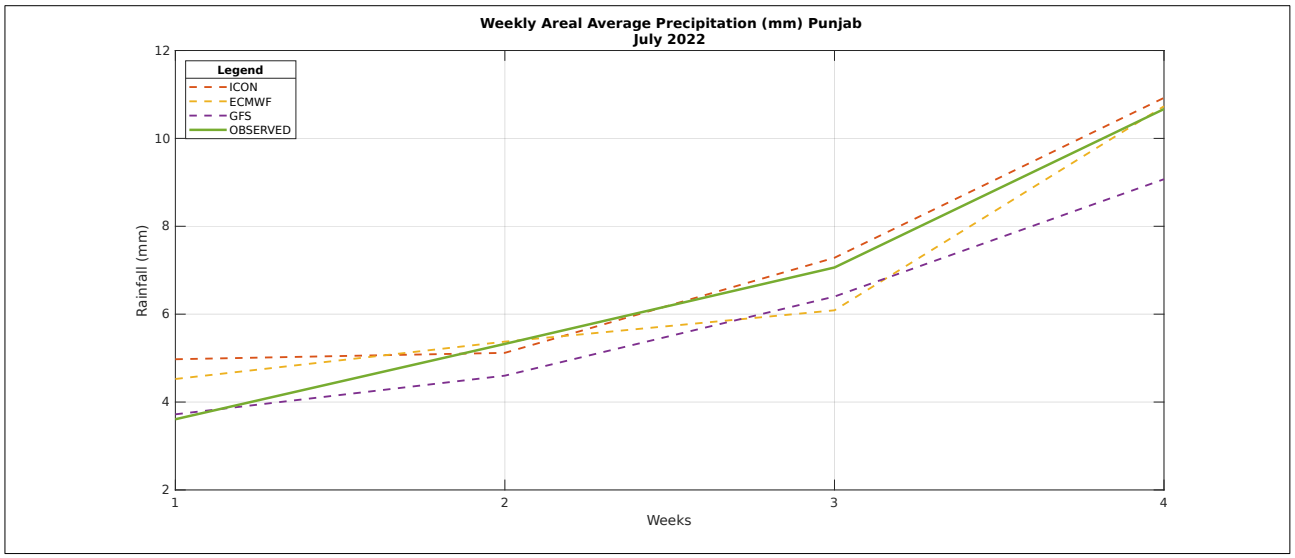


Figure 2.3.7, Weekly Areal Average Precipitation Punjab, Pakistan July 2022

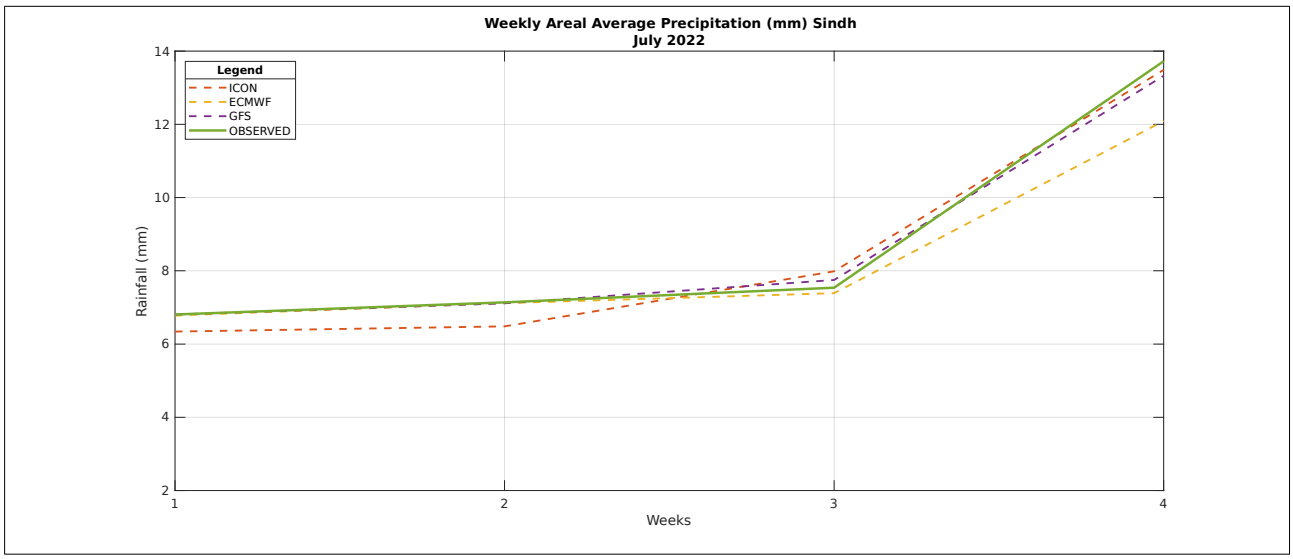


Figure 2.3.8, Weekly Areal Average Precipitation Sindh, Pakistan July 2022

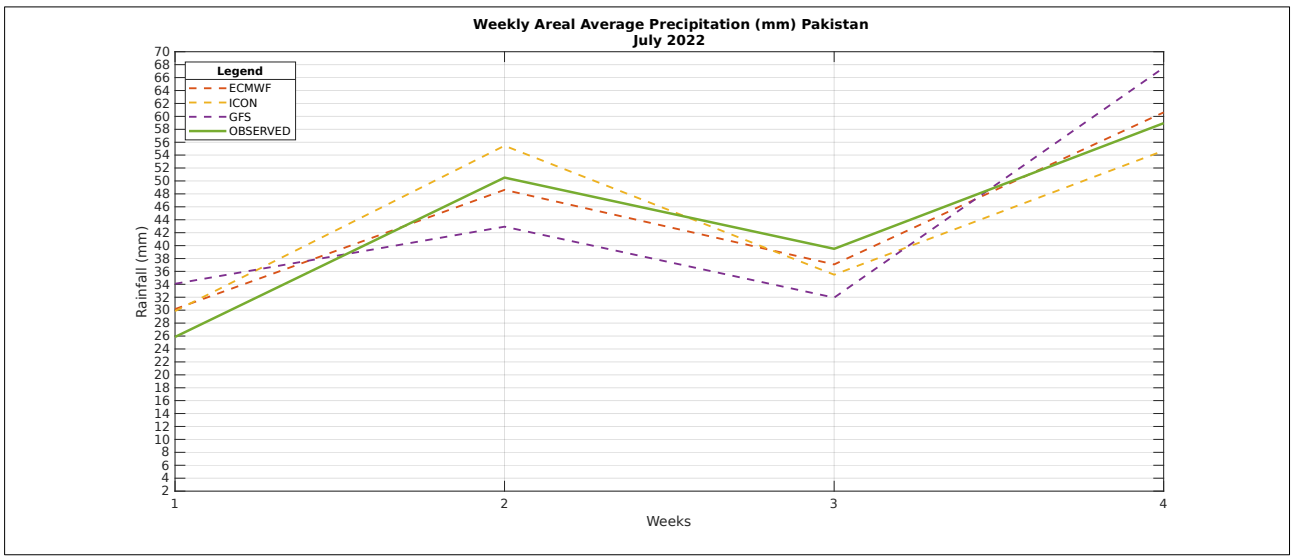


Figure 2.3.9, Weekly Areal Average Precipitation Pakistan July 2022

2.4 Accuracy Assessment of Forecast Models with Observed Data

The previous results do help in analyzing but do not provide any statistics on the degree of accuracy of the models. The following figures give us a coherent picture of the accuracy of the models,

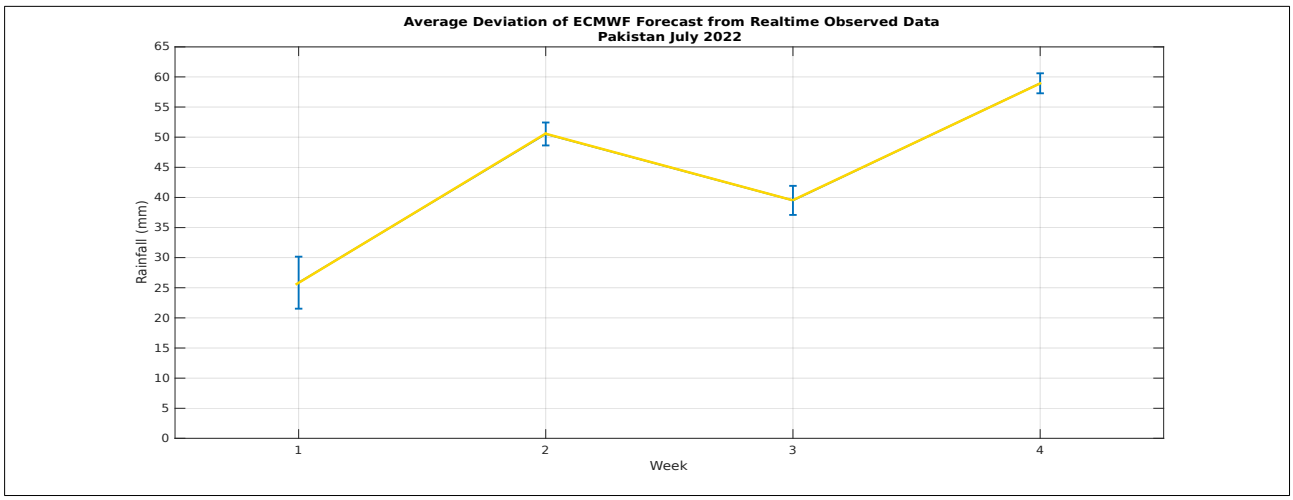


Figure 2.4.1, Deviation Analysis ECMWF

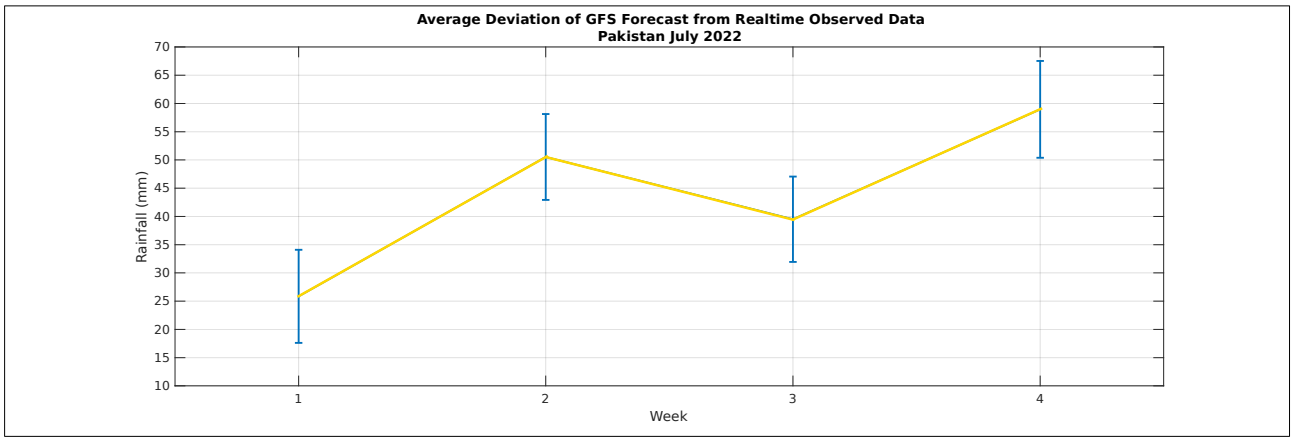


Figure 2.4.2, Deviation Analysis GFS

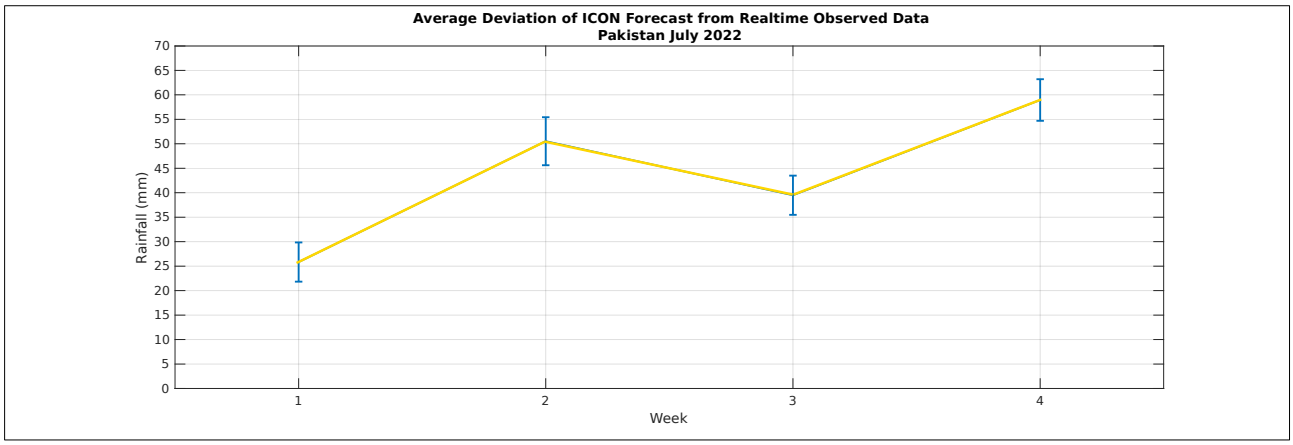


Figure 2.4.3, Deviation Analysis ICON

The above accuracy assessment plots explain better the comparison of the obtained results with each model respectively. It can be deduced that the forecast models lie within ±20% of the actual observations and can be termed as reliable. The casual shifts of peaks can be attributed to wrong classification of forecast models or ill-calibration of timesteps whether by default from the backend forecast stations or during the preprocessing step by the internees. However, from the above figures, ECMWF shows less deviation in comparison and gives more reliable results when plotting numerical values of average precipitation for the month of July 2022.

3 Conclusion

From the above analysis we can see that all provinces of Pakistan has received rainfall with an average lower threshold starting from 20 mm and that many parts of Balochistan and Sindh has persistently received heavy rainfall in the month of July and thus are at a higher risk of floods and thereby explains why these areas are the most impacted by the floods. Punjab, especially southern and eastern Punjab has also received heavy showers however over the span of four weeks, there is greater variation in rainfall distribution.

Heavy monsoon rainfall, rising temperatures due to global warming and melting of glaciers in northern parts are all factors that has led to the prevailing disastrous flooding that has taken hold of one third of the country. The above discussed models along with observed precipitation data of July 2022 clearly indicates that the average rainfall distribution in Pakistan is excessive in many parts and is most likely to trigger floods.